

# Prashik Deshpande

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## Education

| Year | Degree                                       | Institute                                | Score  |
|------|----------------------------------------------|------------------------------------------|--------|
| 2023 | Integrated M.S. in Mathematics and Computing | Indian Institute of Technology Kharagpur | 70.3%  |
| 2017 | Higher Secondary Certificate (HSC)           | St. Paul Junior College                  | 90.31% |
| 2015 | Secondary School Certificate (SSC)           | R. N. T. High School                     | 93.20% |

## Work Experience

**Platform Application Services · Squarepoint Capital · Systems Specialist · Bengaluru** Jul 2023 – Present

### • PAS-Data

Jul 2023 – Apr 2024

- Managed legacy **Chef** cookbooks for **PostgreSQL** configuration management
- Developed custom **Prometheus** exporter and cookbook for collecting ethtool metrics for colocation hosts
- Led initiative to standardize **Redis** deployments using custom **Ansible** collection built from ground up
- Supported L2 and L3 level escalations during on-call duties for PAS and actively resolved alerts and prepared incident reports

### • PAS-Middleware

May 2024 – Oct 2024

- Developed custom health checks for **ClickHouse** clusters in **Consul**
- Enhanced firm-wide **PostgreSQL** inventory collection with AWX scheduled jobs
- Assisted in database migrations from legacy **CentOS** PostgreSQL older versions to newer **Rocky Linux** versions
- Built geo-separated **MS SQL Server** cluster for quantitative research & production data segregation

### • PAS-Databases

Nov 2024 – Present

- Implemented **pgBackRest** for PostgreSQL backups/restores with **Google Cloud Storage** integration
- Stabilized **ClickHouse** backup pipeline with improved scripts and monitoring for duration and failures

## Projects

**GenAI Trip Planner** Hack2Skill AI Hackathon

Sep 2024 – Oct 2024

- Built MCP server in **Python** for personalized trip planning based on user inputs
- Containerized application using Docker, Flask & uv for easy deployment on GCP VMs
- Integrated Google Gemini APIs and Vertex AI for response refinement using dates, locations & travel themes

**Topological Data Analysis and Simplicial Neural Networks** Masters Thesis Project

Jul 2022 – Apr 2023

- Applied persistent homology (topological data analysis) to study shape properties of data under Prof. Bibhas Adhikari
- Extended graph representation learning to simplicial complexes to capture higher-dimensional relations
- Reformulated link prediction as simplex prediction problem for complex networks
- Solved using simplicial neural networks with topological features (supervised setting)

## Positions of Responsibility

**Technology Coordinator, Technology Students' Gymkhana**

Jun 2019 – Feb 2020

- Maintained & improved mobile experience of Gymkhana website for ~15,000 students
- Automated mess menu fetching using Google Sheets API & Fetch API
- Revamped interactive infographics for General Championships, Open IIT & Inter IIT results
- Implemented student ERP authentication for Mess Rebate Form + SQLite-backed portal

## Achievements

- Kishore Vaigyanik Protsahan Yojana (**KVPY**) Scholarship awarded for full course duration (AIR 7)

## Skills & Expertise

|                                       |                                           |
|---------------------------------------|-------------------------------------------|
| <b>Programming Languages</b>          | Python • JavaScript • Ruby • C++          |
| <b>Configuration &amp; IaC</b>        | Ansible • Chef • Terraform                |
| <b>Observability &amp; Monitoring</b> | Prometheus • Grafana • Elasticsearch      |
| <b>Containers &amp; Orchestration</b> | Kubernetes • Docker                       |
| <b>Databases &amp; Storage</b>        | PostgreSQL • Microsoft SQL Server • Redis |
| <b>Cloud Platforms</b>                | AWS • Google Cloud Platform (GCP)         |
| <b>Operating Systems</b>              | Linux • Windows Server Administration     |